

Auto Ticket Classification using Flow Designer

Final Project Report

Project Documentation | Date: 01 October 2026 | Team: Not specified in project brief

1. INTRODUCTION

1.1 Project Overview

The project automates school IT helpdesk ticket classification using ServiceNow Flow Designer. Students and teachers provide a caller and short description; the system creates a structured Incident Workflow record, evaluates keywords, assigns Category and Subcategory, and sends an acknowledgement email.

1.2 Purpose

The purpose is to reduce manual categorization, improve routing consistency, reduce response friction, and provide a maintainable no-code solution that can scale to more issue types.

2. IDEATION PHASE

2.1 Problem Statement

The school IT helpdesk receives recurring Wi-Fi, projector, password, and slow-computer requests. Manual review consumes agent time, creates classification inconsistency, and delays routing.

2.2 Empathy Map Canvas

Students and teachers want a quick, low-effort submission and confirmation. IT agents want standardized fields and correctly routed work. Administrators want maintainable rules and controlled access.

2.3 Brainstorming

The selected idea was deterministic keyword classification with dependent choice logic, a structured custom table, email acknowledgement, update-set portability, and validation scenarios.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

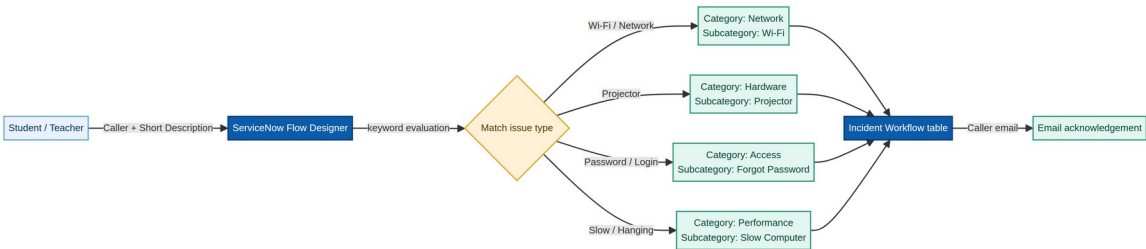
The journey is: issue noticed → description submitted → record created → Flow Designer classifies → ticket routed → caller receives confirmation → agent resolves ticket.

3.2 Solution Requirements

ID	Requirement	Outcome
FR-1	Capture ticket and caller data	Structured Incident Workflow record
FR-2	Classify by keywords	Category and Subcategory assigned automatically
FR-3	Use dependent choices	Valid subcategories mapped to categories
FR-4	Notify caller	Email acknowledgement after creation
NFR-1	Maintainability and scalability	No-code configurable rules and update sets

3.3 Data Flow Diagram

Caller input flows into the Incident Workflow record, then through Flow Designer keyword evaluation, field updates, routing data, and caller email notification.



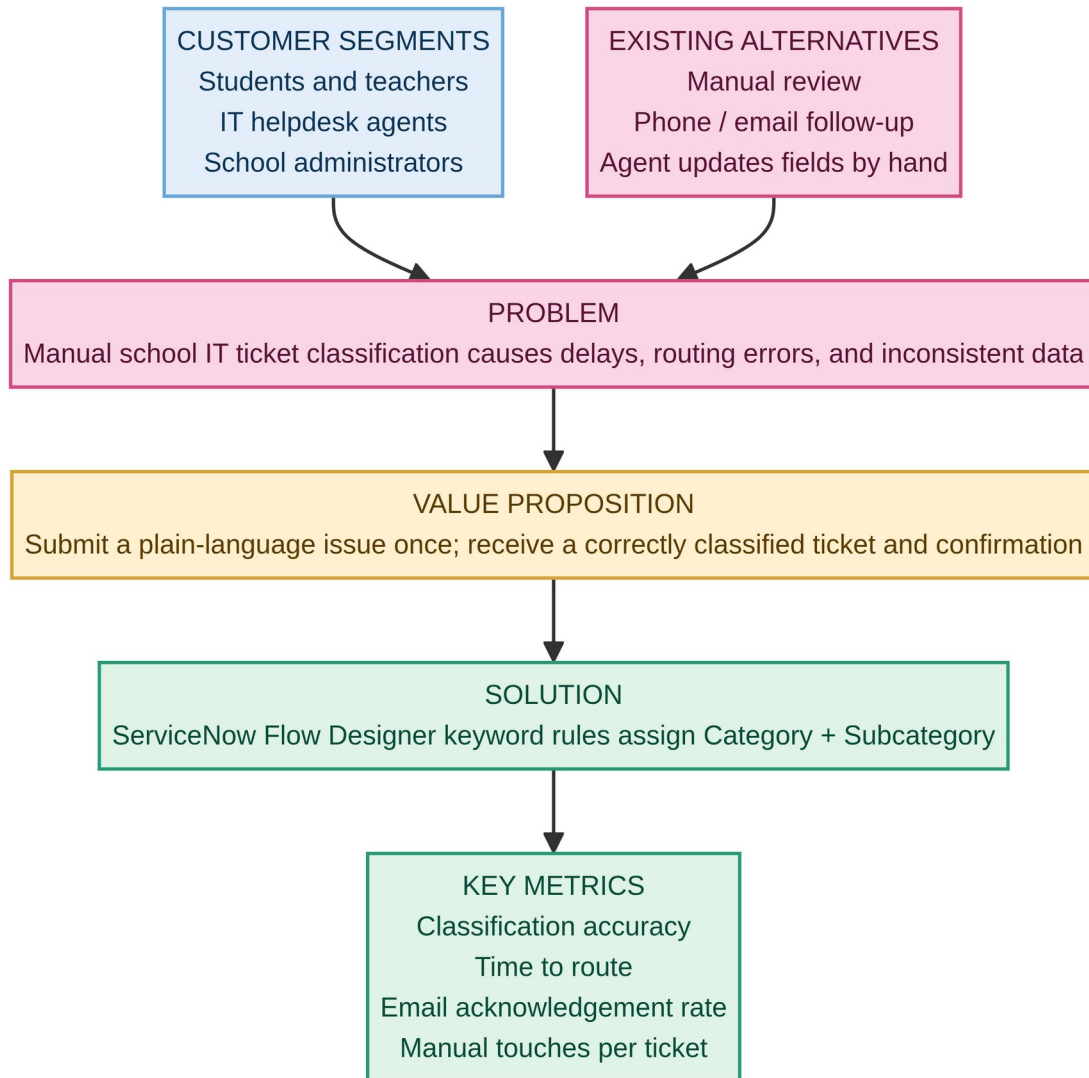
3.4 Technology Stack

Platform: ServiceNow. Automation: Flow Designer. Data: Incident Workflow custom table with reference and choice fields. Deployment: Project Update Set and XML export. Notification: ServiceNow email action. No external API or machine-learning model is required.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

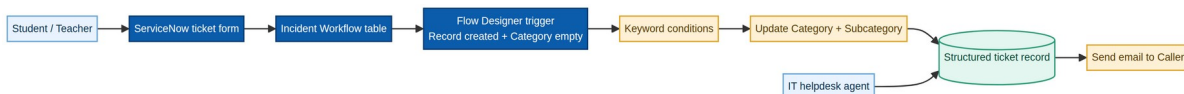
The solution fits the existing behavior because the caller uses the familiar ticket form and email channel while the helpdesk gains automated classification and standardization.



4.2 Proposed Solution

Create the table and fields, configure Category/Subcategory dependencies, activate the “Auto Classify School IT Tickets” flow, update records based on WiFi/Projector/Password/Login/Slow/Hanging keywords, and send the confirmation email.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Three sprints were planned: Sprint 1 requirements and data model (11 points), Sprint 2 automation and communication (7 points), and Sprint 3 validation and deployment (5 points). Total planned effort is 23 story points and planned average velocity is 7.67 points per sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional tests cover ticket intake, four keyword classifications, dependent choices, state values, and email notification. Performance tests cover classification response time, controlled burst creation, duplicate auto-number checks, flow failures, and notification throughput. The supplied test plans identify expected outcomes and require actual evidence from the ServiceNow environment.

Test ID	Scenario	Expected result
FT-02	Wi-Fi keyword	Network / Wi-Fi
FT-03	Projector keyword	Hardware / Projector
FT-04	Password/Login keyword	Access / Forgot Password
FT-05	Slow/Hanging keyword	Performance / Slow Computer
PT-01	Response time	Within agreed helpdesk threshold

7. RESULTS

7.1 Output Evidence

The planned output is a classified Incident Workflow record and a caller email acknowledgement. Evidence should include the saved record, field values, flow execution, email log, and test result for each scenario. No unverified execution result is claimed in this report.

8. ADVANTAGES & DISADVANTAGES

Advantages: no-code maintenance, consistent classification, lower manual effort, faster routing, dependent data quality, email confirmation, update-set portability. Disadvantages: keyword rules may miss synonyms or ambiguous text, classification depends on accurate descriptions, and additional issue types require rule maintenance.

9. CONCLUSION

The project delivers an end-to-end ServiceNow automation design for school IT support. It addresses the manual classification problem through structured records, deterministic Flow Designer rules, dependent choices, notification, testing, and deployment documentation.

10. FUTURE SCOPE

Add fallback/manual-review routing, synonym dictionaries, dashboards and SLA analytics, additional issue categories, richer audit history, multilingual keywords, and integration with service catalogs or knowledge articles.

11. APPENDIX

Source code: No custom source code; configuration is maintained in ServiceNow and captured in the Project Update Set. Dataset link: Not applicable; test data consists of controlled school IT ticket descriptions. GitHub / demo link: Not provided. Related deliverables: Requirement Analysis, Project Design, Project Planning, UAT, and Functional/Performance Testing documents.